preface

My journey into the world of intelligent systems began back in the early 1980s. Like many people then, I believed artificial intelligence (AI) was just around the corner. It always seemed like one more innovation and technological leap would lead us to the intelligence we imagined. But that leap never came.

Perhaps the promise of HAL, from Stanley Kubrick's *2001: A Space Odyssey*, captivated me with the idea of a truly intelligent computer companion. After years of effort, trial, and countless errors, I began to understand that creating AI was far more complex than we humans had imagined. In the early 1990s, I shifted my focus, applying my skills to more tangible goals in other industries.

Not until the late 1990s, after experiencing a series of challenging and transformative events, did I realize my passion for building intelligent systems. I knew these systems might never reach the superintelligence of HAL, but I was okay with that. I found fulfillment in working with machine learning and data science, creating models that could learn and adapt. For more than 20 years, I thrived in this space, tackling problems that required creativity, precision, and a sense of possibility.

During that time, I worked on everything from genetic algorithms for predicting unknown inputs to developing generative learning models for horizontal drilling in the oil-and-gas sector. These experiences led me to write, where I shared my knowledge by way of books on various topics—reverse-engineering Pokémon Go, building augmented and virtual reality experiences, designing audio for games, and applying reinforcement learning to create intelligent agents. I spent years knuckles-deep in code, developing agents in Unity ML-Agents and deep reinforcement learning.

Even then, I never imagined that one day I could simply describe what I wanted to an AI model, and it would make it happen. I never imagined that, in my lifetime, I would be able to collaborate with an AI as naturally